SFID: An Azimuthal Structure-Guided Robust and Adaptive Interference Detector for SAR Imagery

Lunhao Duan, Xingyu Lu, *Member, IEEE*, Jianchao Yang, Huizhang Yang, and Hong Gu

***Abstract*—Synthetic aperture radar (SAR) is highly susceptible to radio frequency interference (RFI). A novel RFI detector, the singular value decomposition-fractional Fourier transform interference detector (SFID), based on left singular vector characteristics, is proposed for single-look complex (SLC) SAR images. The core principle is to exploit the azimuthal structure difference between focused targets and interference: the target response manifests as a sinc function, while the interference response exhibits a linear frequency-modulated (LFM) form. Specifically, the method segments images into patches, performs the singular value decomposition (SVD) on each patch, applies the fractional Fourier transform (FRFT) at the optimal angle to the left singular vector matrix, and realizes constant false alarm rate (CFAR) detection by comparing the squared global FRFT peak, p_1, with an adaptive threshold. Theoretical derivation shows that p_1 obeys the Gumbel distribution in clean SAR images, providing a basis for adaptive threshold calculation. For the terrain observation by progressive scans (TOPS) mode, the linear variation of the focused target Doppler bandwidth centroid degrades detection performance; thus, an interference model for the TOPS mode is established, and detection preprocessing tailored to this mode is introduced to enhance detection robustness. Experiments on measured data from Gaofen-3 and Sentinel-1 demonstrate that SFID can effectively detect low interference-to-signal ratio (ISR) and dense interference, outperforming traditional detection methods.**

***Index Terms*—Fractional Fourier transform (FRFT), interference detection, singular value decomposition (SVD), synthetic aperture radar (SAR), terrain observation by progressive scans (TOPS).**

I. Introduction

A. Background

SYNTHETIC aperture radar (SAR) is an active microwave imaging system. Its all-weather, all-time imaging capability and high-resolution image quality make it a powerful remote sensing tool, which has been widely applied in areas such as remote sensing monitoring and disaster assessment [1], [2]. However, due to the overlap in spectral resources, when SAR is operating, in addition to the target echo, it can easily receive signals from other cofrequency devices; i.e., it suffers from radio frequency interference (RFI), such as from ground radars, communication equipment, or other electronic devices [3], [4]. Since these signals generally do not match the imaging operator in either the azimuth or the range direction, they typically appear as bright stripes in the image, as shown in Fig. 1(a). This severely degrades SAR image quality and impacts subsequent SAR image applications. Therefore, researchers have proposed a series of interference suppression methods [5], [6], [7]. To avoid unnecessary loss of targets, the positions of interference stripes in the image need to be obtained in advance. Hence, effective interference detection for SAR images is of great practical significance. Moreover, some interference source localization methods [8] and image quality assessment techniques [9] also require the acquisition of interference stripe positions. To this end, this article conducts research on SAR RFI detection.

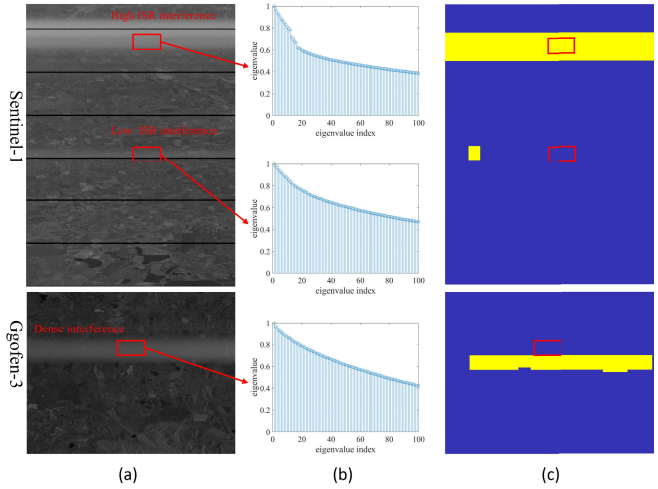


Fig. 1. Performance analysis of the Lambda-1 detector. (a) S-1 and GF-3 images. (b) First 100 eigenvalues. (c) Detection results, where yellow indicates interference presence.

B. Existing Works

Currently, interference detection methods can be mainly divided into three categories, one of which is based on traditional signal processing techniques. Leng et al. [10] leveraged the RFI characteristics in dual-polarization GRD images and proposed an RFI index for RFI detection. Lv et al. [11] introduced a new RFI detector based on the combination of eigenvalue decomposition (EVD) and short-time Fourier transform (STFT). Li et al. [12] used range spectrum differences

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to capture the characteristics of mutual RFI and employed the k-means algorithm for interference detection. Bollian et al. [13] achieved interference detection using time-bandpass filters. Fu et al. [14] accomplished detection by weighted fusion of time-domain kurtosis and time-domain first-order partial derivatives. Yang et al. [15] utilized subband spectral kurtosis to detect RFI. Huang et al. [16] developed a method called slow-time eigenvalue slope thresholding. Natsuaki et al. [17] identified interference in the time and frequency domains of raw data by analyzing interferograms between different receiving channels. Monti-Guarnieriet al. [18] leveraged the range spectrum characteristics of interference, combined with a one-tailed Fisher's Z-test for detection.

The other category is based on deep learning methods. Machine learning has become increasingly extensive across various application fields, including detection [19], [20], [21]. Zhao et al. [22] proposed the IQDN algorithm, designed based on a modified MobileNet V2 network, which can effectively detect RFI in the image domain. Tao et al. [23] adopted an encoder–decoder architecture, incorporating a complex spatial pyramid pooling module, depthwise convolution, and a self-attention mechanism to propose a semantic cognitive enhancement network. Cen et al. [24] proposed a convolutional autoencoder deep learning model for interference detection. Chojka et al. [25] utilized techniques including pixel convolution, thresholding, and nearest-neighbor structural filtering for feature extraction to achieve rapid detection in image viewing. Tao et al. [26] proposed a method for detecting RFI signals in SAR raw data using time–frequency semantic analysis. Li et al. [27] proposed a semantic cognitive enhancement network for RFI detection in Sentinel-1 SAR images. Yu et al. [28], Fan et al. [3], and [29] used STFT to obtain the time–frequency spectrum of echoes and trained networks capable of detecting interference using time–frequency datasets.

Another category is based on time series analysis methods. Tao et al. [30] leveraged the coupling correlation and complementary information between time-series images to propose an interference extraction scheme for SAR images using multitemporal analysis. Li et al. [31] utilized previous time-series SAR images to construct an approximately RFI-free background image, generated a difference image based on change detection methods, and performed analysis using an adaptive threshold to achieve interference detection.

In summary, traditional signal processing-based methods are mostly only applicable to the echo domain or dual-polarization images. Furthermore, many methods are only suitable for specific types of interference, such as narrowband interference (NBI). Time-series-based methods rely on previously acquired data and cannot be applied to single-pass SAR data acquisitions. Deep learning-based methods generally depend on large-scale datasets for training [32], [33]. Moreover, they also face certain challenges in generalizing across different SAR systems [9].

To overcome the aforementioned issues, Yang et al. [9] proposed an interference detection method based on the maximum eigenvalue of the image. In SAR images, the interference response after imaging exhibits a rank-1 property, meaning that interfered images generally possess a dominant eigenvalue. Moreover, the maximum eigenvalue of clean images approximately follows the Tracy–Widom distribution of order 2. Thus, by performing EVD on image patches and comparing the largest eigenvalue with an adaptive threshold, effective detection of interfered images can be achieved. The method exhibits high detection efficiency, delivers satisfactory detection performance for both narrowband and wideband interference, and applies to various SAR systems.

C. Limitations of Existing Methods

However, since the power of some interference is not much greater than that of the target, existing methods still have certain limitations. In our work on SAR image interference detection, we have found that the Lambda-1 detector [9] is inadequate for effectively detecting the following two types of interference. The first is low interference-to-signal ratio (ISR) interference, where the relatively weak energy of the interference results in a less prominent corresponding eigenvalue. The second is dense interference, which refers in this article to interference signals that are highly concentrated in the echo domain, even to the extent of being linear frequency-modulated continuous wave (LFMCW) signals that persist across hundreds of pulse repetition interval (PRI) cycles. During the imaging process, interference originally located within a single slow-time interval manifests in the image domain as rectangular stripe patterns spanning multiple PRI cycles.

When numerous interference pulses are present in the echo domain, the interference fringes overlap, resulting in extremely high energy in the superimposed regions. After performing EVD, the interference energy is not concentrated solely in the first or a few eigenvalues but is distributed across more eigenvalues. Furthermore, since the Lambda-1 detector's threshold is related to the variance of the interfered image, the high energy elevates the threshold, thereby causing the first eigenvalue to also appear less prominent. In summary, the reason why these two types of interference are difficult to detect lies in the fact that the first eigenvalue is not obvious. Low ISR interference has an eigenvalue that is inherently not prominent, while dense interference is not significant relative to the entire image.

To examine these two cases, we analyzed measured SAR images. Fig. 1(a) shows two SAR images affected by interference. The first one is from Sentinel-1, which contains one weak interference and one strong interference. The other is from Gaofen-3, where the interference is an LFMCW signal that persists over multiple PRI periods in the echo domain, representing a typical case of dense interference. By performing EVD on the image patch within the red rectangle in Fig. 1(a), we obtain the results shown in Fig. 1(b). Consistent with the above analysis, the first eigenvalue does not exhibit distinct prominence in patches with low-ISR interference and dense interference, which explains why the Lambda-1 detector fails to achieve reliable detection of such interference. Fig. 1(c) shows the detection results for the entire image, with yellow indicating the presence of interference. Detailed interference

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detection experiments on these two images will be presented in Section VI.

D. Solution and Contributions of This Article

In this article, we propose a new SAR image interference detector, termed the SVD-FRFT interference detector (SFID). Compared with the traditional Lambda-1 detector, the proposed SFID can effectively handle SAR images affected by low ISR interference or dense interference. SAR images may contain target scatterers, speckle noise, RFI, and other background signals.

The unique nature of RFI stems from its incoherence with the SAR imaging operator. In the SAR echo matrix, a single RFI pulse typically appears as a δ function in the azimuth direction. Consequently, after focusing, the interference stripes exhibit an azimuthal structure consistent with the azimuth matched filter. For example, in a linear frequency modulation (LFM)-based SAR system, the interference stripes clearly manifest as LFM signals in the azimuth direction. In contrast, target echoes are coherently matched with the imaging operator in both range and azimuth, and after imaging, they appear as a deterministic point spread function (usually a 2-D sinc function). Similarly, speckle noise, which is a multiplicative noise generated by the coherent superposition of echoes from a large number of randomly distributed scatterers within a single resolution cell, also undergoes the same 2-D coherent matching process. Therefore, the azimuthal profile of speckle does not exhibit an LFM structure either.

In summary, the unique deterministic azimuthal structure of RFI, which matches the imaging operator, provides a robust feature that can effectively distinguish it from other signals. This fundamental difference lays the theoretical foundation for subsequent RFI detection.

The core idea of the proposed method is that after performing singular value decomposition (SVD) on an SAR image patch, the azimuth singular vector corresponding to the interference component exhibits LFM characteristics, leading to anomalous peaks in the fractional Fourier transform (FRFT) domain, which enables interference detection. Specifically, according to SAR imaging principles and SVD theory, although the first few eigenvalues corresponding to the aforementioned interference types may not be prominent, their azimuth singular vectors still retain LFM properties. Leveraging this property, we first divide the SAR image into patches and perform SVD on each patch. Then, we perform the FRFT at the optimal angle on the azimuth singular vectors to obtain p_1, the square of the global maximum FRFT peak. Finally, we compare p_1 with a threshold to complete interference detection. Based on the theoretical derivation, p_1 obtained from clean SAR images follows a Gumbel distribution; thus, the threshold can be adaptively determined according to this distribution. Furthermore, according to the imaging principles, the optimal angle can be directly computed from the SAR system parameters.

The main contributions of this article are summarized as follows.

1) *SFID Detector:* We propose a new interference detection framework based on the observation that the squared FRFT peaks of SAR image singular vector matrices follow a Gumbel distribution. Compared to traditional SAR image interference detection methods, the proposed method effectively handles low ISR interference and dense interference. Furthermore, it offers high processing efficiency and is capable of providing more reliable results for downstream tasks such as interference suppression or interference parameter estimation.
2) *Interference Modeling and Detection Preprocessing in TOPS Mode:* Compared to other imaging modes, the signal model in terrain observation by progressive scans (TOPS) mode is more complex. We analyze the target and interference models in focused images under this mode, as well as their impact on interference detection, and introduce a preprocessing scheme suitable for this mode to enhance detection robustness.
3) *Comprehensive Experimental Validation:* We conduct extensive experiments using measured Gaofen-3 and Sentinel-1 single-look complex (SLC) images. The detection performance is quantitatively evaluated using metrics such as the false alarm probability (P_{fa}) and detection probability (P_d). The results demonstrate the effectiveness of the proposed detector.

E. Organization of This Article

The remainder of this article is organized as follows. Section II introduces the signal model in SAR images. Section III presents the proposed detection method. Section IV discusses the impact of the TOPS mode on the detection method. Section V provides the detection flow. Section VI analyzes the experimental results. Finally, Section VII concludes this article.

II. Signal Model in SAR Image

This section aims to establish a mathematical model for target and interference in the focused image, clarifying the physical meanings represented by the singular values and singular vectors of the interference after SVD, thereby providing a theoretical foundation for subsequent interference detection. Since most existing SAR systems employ an LFM signal as a transmitted signal, the detection method proposed in this article is specifically designed under this condition.

A. Target Model

Typical imaging algorithms such as Chirp Scaling, Range–Doppler, and Omega-K primarily consist of range matched filtering, range cell migration correction (RCMC), and azimuth matched filtering. The main role of RCMC is to correct the target echo offset caused by SAR platform motion, without fundamentally altering the basic form of the signal in the final imaging result [34]. Therefore, the imaging process can be approximated as a 2-D matched filtering operation in range and azimuth [5]. Since this article assumes that the SAR system employs an LFM transmit signal, the range matched filtering essentially corresponds to the convolution of two coherent LFM signals, which can be approximated as a sinc function. Moreover, due to the relative motion between the

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platform and the target, the variation in slant range leads to a linear relationship between the generated Doppler frequency and the azimuth time. Consequently, the target echo also exhibits an LFM characteristic in the azimuth direction, with its chirp rate k_a defined as

$$k_a = -\frac{2f_0 V_r^2}{cR} \tag{1}$$

where V_r is the effective radar velocity, c is the light velocity, f_0 is the system frequency, and R is the slant range.

Therefore, the azimuth processing is also the convolution of the LFM signal. The focused target model has been provided [34] and can be expressed as the following range–azimuth decoupled model:

$$T(\eta;\tau) = \sigma \cdot a(\eta) \cdot r(\tau) \tag{2}$$

where σ denotes the coefficient, $a(\cdot)$ and $r(\cdot)$ denote the azimuth and range matching outputs, typically as sinc function.

B. Interference Model

Considering a single interference pulse, in the 2-D echo matrix, it can be characterized as the following range–azimuth decoupled model:

$$j(\eta;\tau) = e_a(\eta)\, e_r(\tau) = \delta(\eta - \eta_0)\, e_r(\tau) \tag{3}$$

where e_a is the azimuth function (which is a δ function), e_r is the range function, and η_0 is the azimuth time of the interference pulse.

The imaging process can be expressed as follows [5]:

$$J(\eta';\tau') = \int\int e_a(\eta)\, e_r(\tau) \cdot h(\eta,\tau;\eta',\tau')\, d\eta d\tau \tag{4}$$

where h denotes the 2-D convolution kernel, and $(\eta';\tau')$ is the matched time. Applying an order-N Taylor approximation to the kernel

$$h(\eta,\tau;\eta',\tau') \approx \sum_{n=1}^{N} u_n(\eta-\eta')\, v_n(\tau-\tau') \tag{5}$$

where $u(\eta)$ and $v(\tau)$ are the azimuth and range function of the kernel. Both (3) and (5) can be expressed as 2-D decoupled functions, (4) can be further expressed as

$$J(\eta';\tau') \approx \sum_{n=1}^{N} a_n(\eta')\, r_n(\tau') \tag{6}$$

where $a(\eta')$ and $r(\tau')$ are the convolution outputs in the azimuth and range directions, respectively, and can be expressed as

$$\begin{aligned} a_n(\eta') &\approx \mathrm{conv}(u_n(\eta), e_a(\eta)) \\ &= \mathrm{conv}(u_n(\eta), \delta(\eta-\eta_0)) = u_n(\eta') \\ r_n(\tau') &\approx \mathrm{conv}(v_n(\tau), e_r(\tau)) \end{aligned} \tag{7}$$

where conv(·) is the convolution function.

Since the target echo exhibits an LFM form in both the range and azimuth directions, the convolution kernels $v_n(\cdot)$ and $u_n(\cdot)$ are both LFM signals. For a single interference pulse, its range form $e_r(\tau)$ depends on the type of interference signal, such as common LFM, single-frequency, or noise-frequency modulation interference. Therefore, for different interference types or parameters, the output of $r(\tau')$ is not fixed, making it difficult to establish a universal interference detection model. However, in the 2-D echo matrix, a single interference pulse appears as a δ function in the azimuth direction, independent of parameters such as interference type. After imaging, the azimuth output $a_n(\eta')$ can be expressed as the convolution of $u_n(\eta)$ and the δ function, maintaining the same form as $u_n(\cdot)$, i.e., an LFM signal with a chirp rate of $-k_a$. Therefore, we can utilize the azimuth output to establish a universal interference detection model.

Since both the target response in (2) and the interference response in (6) can be expressed in the form of a 2-D decoupled product in range and azimuth, we can rewrite (2) or (6) using eigenvalues and singular vectors

$$\begin{aligned} T(\eta;\tau) &\approx \sqrt{\lambda} \cdot p(\eta) \cdot q(\tau) \\ J(\eta';\tau') &\approx \sum_{k=1}^{K_j} \sqrt{\lambda_k} \cdot p_k(\eta') \cdot q_k(\tau') \end{aligned} \tag{8}$$

where p and q denote the left and right singular vectors, λ is the eigenvalue. Due to processes such as RCMC, the interference response is typically distributed across K_j singular vectors.

C. Analysis of Left Singular Vectors for SLC Images

Eigenvalues reflect the energy of corresponding components, with larger values indicating stronger energy. Therefore, reference [9] effectively detects strong interference by leveraging this property. Low ISR interference possesses weak energy, which is comparable to that of the background targets, and thus, it is difficult to cause an anomaly in the first eigenvalue, thereby failing to effectively detect such interference.

Left singular vectors represent the azimuthal structure of the components. The left singular vectors $p(\cdot)$ of the target are sinc functions, while those of the interference are LFM signals. No matter the type of interference, as long as interference is present, its corresponding left singular vectors retain the LFM characteristic, and the chirp rate is solely determined by the system parameters. Therefore, we can focus this LFM signal via FRFT at the optimal angle to generate distinct FRFT peaks. In contrast, the left singular vectors of clean images do not exhibit the LFM characteristic, and applying FRFT to them yields no anomalous peaks.

This allows us to create a clear discrepancy between interfered and clean images, thereby enabling effective interference detection. Based on this observation, we will develop a statistical detection method that leverages this distinction within the singular vectors.

III. SFID for SAR Image Interference Detection

In this section, we propose a new framework for adaptive interference detection in SAR images.

A. Binary Hypothesis Testing Problem

The interference detection in SAR images can be formulated as a binary hypothesis testing problem.

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1) H_0: The image consists only of ground targets and noise, with no interference stripes.
2) H_1: The image contains interference stripes.

Under the H_0 hypothesis, when an SAR image contains no interference stripes, its azimuth singular vectors do not exhibit LFM characteristics, and the peaks p_1 after FRFT should remain stable without showing anomalous values. Therefore, we first theoretically derive the statistical distribution of p_1 under the H_0 hypothesis, which allows us to select an appropriate threshold for a given false alarm probability. Interference detection is then completed by comparing p_1 with this threshold.

B. Distribution Characteristics of the Squared Maximum Value in SAR Image Singular Vector Matrices

Assume an SAR image patch $\mathbf{S}$ of size $N_\mathrm{a} \times N_\mathrm{r}$, where each element in the patch is independent and identically distributed (i.i.d.) and approximately follows a zero-mean complex Gaussian distribution [9]. First, perform SVD on it

$$\mathbf{S} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathrm{H}} \tag{9}$$

where $\mathbf{U} \in \mathbb{C}^{N_\mathrm{a}\times N_\mathrm{a}}$ and $\mathbf{V} \in \mathbb{C}^{N_\mathrm{r}\times N_\mathrm{r}}$ are the azimuth (left singular vectors) and range (right singular vectors) singular vector matrices, respectively; $\mathbf{\Sigma} \in \mathbb{R}^{N_\mathrm{a}\times N_\mathrm{r}}$ is a diagonal matrix whose diagonal elements are the singular values. $\mathbf{U}$ can be expressed as follows:

$$\mathbf{U} = \left[u_1 \dots u_i \dots u_{N_\mathrm{a}}\right] \in \mathbb{C}^{N_\mathrm{a}\times N_\mathrm{a}} \tag{10}$$

where each column u_i is an azimuth singular vector. Next, we present two important foundational conclusions [35], [36].

1) After performing SVD on a complex matrix, both $\mathbf{U}$ and $\mathbf{V}$ are unitary matrices.
2) After performing SVD on a matrix composed of i.i.d. zero-mean complex Gaussian random variables, the singular vector matrices follow the Haar measure, i.e., the Haar uniform distribution.

Therefore, the azimuth singular vector matrix $\mathbf{U}$ is distributed according to the Haar measure on the unitary group. In the literature, it is shown that when the dimension of such a matrix is large, the cumulative distribution function (cdf) of its squared maximum value follows a Gumbel distribution [37]. This can be expressed as follows:

$$\lim_{N_a\to\infty} P\left(N_a W_m^2 - 2\ln N_a \le x\right) = \exp\left(-\exp\left(-x\right)\right) \tag{11}$$

for any $x \in \mathbb{R}$. Where $W_m = \max|\mathbf{U}|$ denotes the maximum absolute value of the elements in $\mathbf{U}$.

Although the statistical distribution has been established, interference generally does not induce anomalies in the maximum value of $\mathbf{U}$. Therefore, it is not feasible to directly determine the presence of interference based solely on the maximum value of $\mathbf{U}$. Consequently, we need to identify a transformation that can more effectively capture the differences in the maximum values between the H_0 and H_1 hypotheses while preserving the original statistical distribution properties.

Based on the analysis in Section II, the azimuth singular vectors of interference exhibit an LFM structure, whereas ground targets do not possess this characteristic. Considering that the FRFT not only provides excellent energy concentration for LFM signals but also constitutes a unitary transformation, we apply the FRFT to the matrix $\mathbf{U}$

$$\mathbf{U}' = \mathbf{F}_\alpha \mathbf{U} \tag{12}$$

where $\mathbf{F}_\alpha$ is the FRFT matrix corresponding to the fractional angle α. Because the Haar measure on the unitary group is left-translation invariant [38], and the product of two unitary matrices remains unitary, the transformed matrix $\mathbf{U}'$ remains Haar distributed on the unitary group. Its squared maximum value still follows the distribution shown in (10).

In practical processing, we can first remove strong scatterers from the image. Then, the interference energy is primarily concentrated within the first few singular vectors. Consequently, we may apply the FRFT only to the first K singular vectors and record the maximum absolute value. The maximum absolute value of the ith singular vector after FRFT is defined as

$$P_i = \max\left|\mathbf{F}_\alpha\left(u_i\right)\right| \quad i = 1, \dots, K. \tag{13}$$

The squared global maximum peak extracted from all considered singular vectors is

$$p_1 = \max_{i=1,\dots,K} P_i^2 \tag{14}$$

Then the detection statistic p_1 is compared with a predetermined threshold.

C. Adaptive Threshold Determination

Under the H_0 hypothesis, the cdf of the detection statistic p_1 follows (11). Let $y = (x + 2\ln N_a)/N_a$, we can obtain

$$\begin{aligned} F_{p_1}(y) &= P\left(p_1 \le y\right) \\ &= \exp\left(-\exp\left(-\left(N_a y - 2\ln N_a\right)\right)\right). \end{aligned} \tag{15}$$

Given a false alarm probability P_{fa}, the threshold p_{th} satisfy

$$\begin{aligned} P_{fa} &= P\left(p_1 > p_{th}\right) = 1 - F_{p_1}\left(p_{th}\right) \\ &= 1 - \exp\left(-\exp\left(-\left(N_a p_{th} - 2\ln N_a\right)\right)\right). \end{aligned} \tag{16}$$

Finally, the detection threshold is given by

$$p_{th} = \frac{2\ln N_a - \ln\left(-\ln\left(1 - P_{fa}\right)\right)}{N_a}. \tag{17}$$

In the above derivation, we assume that SAR images follow a Gaussian distribution, leading to the conclusion that $\mathbf{U}$ obeys the Haar measure. However, real SLC images contain strong scatterers, clutter, and textures, which may deviate from the strict Gaussian assumption.

In random matrix theory, the asymptotic Haar measure of the eigenmatrix (i.e., the left singular vector matrix involved in this article) of a large-dimensional sample covariance matrix is a classic conjecture [39]. Early studies [40], [41] pointed out that under an independent component structure, the validity of this conjecture requires the fourth-order moments of the matrix entries to exactly match those of a Gaussian distribution. Subsequent work [42] showed that for elliptically correlated structures with nonlinear dependencies, the asymptotic convergence of the eigenmatrix to the Haar distribution only

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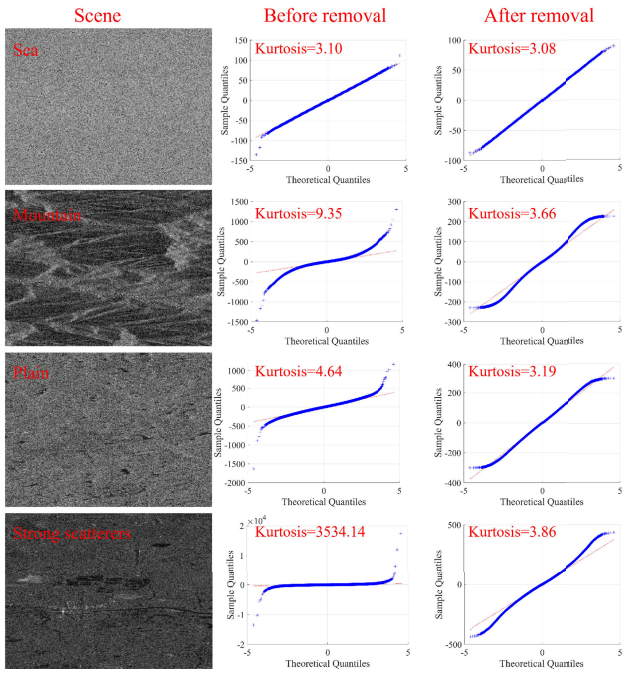


Fig. 2. Analysis of data distribution under different scenarios.

requires a light-tailed condition, without the strict equality of fourth-order moments. More generally, when the data have subexponential tails, the anisotropic local law [43] and the edge universality theory [44] guarantee that both the quadratic forms of the resolvent and the edge eigenvalues obey the same universal limits as in the Gaussian case, and their results can be further used to analyze the distribution of eigenvectors.

Given that strong scatterers in real SAR images induce pronounced heavy-tail effects, thereby violating the aforementioned theoretical assumptions, they should be removed before RFI detection. This preprocessing step allows the data distribution to better approximate the required conditions of fourth-order moments matching or light tails. Fig. 2 shows Q–Q plots and kurtosis (fourth-order moment) for various scenes to assess deviation from a Gaussian distribution. Kurtosis measures tail thickness and peakedness. For a real Gaussian distribution, the kurtosis equals 3.

It can be seen that the sea area, which has no complex texture or large-area strong scatterers, closely follows a Gaussian distribution. Since the plain area contains several strong scatterers, after removal, it also gradually conforms to a Gaussian distribution, and its kurtosis approaches three. The other scenarios seriously deviate from a Gaussian distribution, but after removal, although they still slightly deviate from the Gaussian distribution, their kurtosis gradually approaches 3. Therefore, preprocessing is necessary to remove strong scatterers.

However, the dimension of the patch obtained through segmentation is not infinitely large, which may cause the actual p_1 to be larger than its value under ideal conditions, thereby leading to an increase in false alarms. To address this issue, we need to introduce an additional hyperparameter $\gamma \geq 1$ and use γp_{th} as the actual detection threshold.

If $p_1 > \gamma p_{th}$, it can be concluded that the patch contains interference.

D. Determination of the FRFT Angle α

One of the key points of the proposed method is that FRFT has good energy concentration performance for the LFM signal, so selecting the appropriate FRFT angle is crucial. According to the derivation in Section II, the interference appears as an LFM signal with a chirp rate of $-k_a$ in the azimuth direction.

FRFT can be essentially viewed as a rotation of the coordinate axes in the time–frequency plane. In the time-frequency plane, an LFM signal appears as a tilted straight line, whose slope is exactly the chirp rate. When the rotation angle of the FRFT is such that the new coordinate component is exactly orthogonal to that instantaneous frequency line, the projection of the LFM signal in the fractional Fourier domain converges into an extremely narrow impulse peak, achieving maximum energy concentration. Therefore, we can compute the corresponding rotation angle α based on the interference chirp rate $-k_a$. According to the derivation in [45], we can directly determine this angle as follows:

$$\alpha = \operatorname{arccot}(k_a) \tag{18}$$

where arccot denotes the inverse cotangent function. k_a can be computed via (1) or read directly from the azimuth chirp rate provided in the SLC-product metadata.

IV. Specific Characteristics of Signal Model in TOPS Mode

We have developed an interference detection method. However, during interference detection experiments, we found that this method exhibits extremely unstable detection performance for SLC images acquired by the TOPS mode—such as a high false alarm probability. Therefore, this section will establish an interference model under the TOPS Mode and provide a detection preprocessing scheme for SLC images under this mode.

A. Signal Model Under the TOPS Mode

In traditional imaging modes, the imaging algorithms mainly include 2-D focusing and RCMC. However, in the TOPS mode, the target bandwidth received by the system is much larger than the system pulse repetition frequency (PRF), resulting in aliasing in the frequency domain [46]. To enable precise imaging, the azimuth preprocessing step must be applied.

Azimuth preprocessing ensures that the data meet the requirements for subsequent azimuth compression and RCMC. The main steps include spectrum mosaicking, deramping, low-pass filtering (LPF), and reramping. Assuming the total bandwidth is B_T, aliasing occurs M_s times in the azimuth frequency domain

$$M_s = \left\lceil \frac{B_T}{f_a} \right\rceil = \left\lceil \frac{k_r T_B}{f_a} \right\rceil \tag{19}$$

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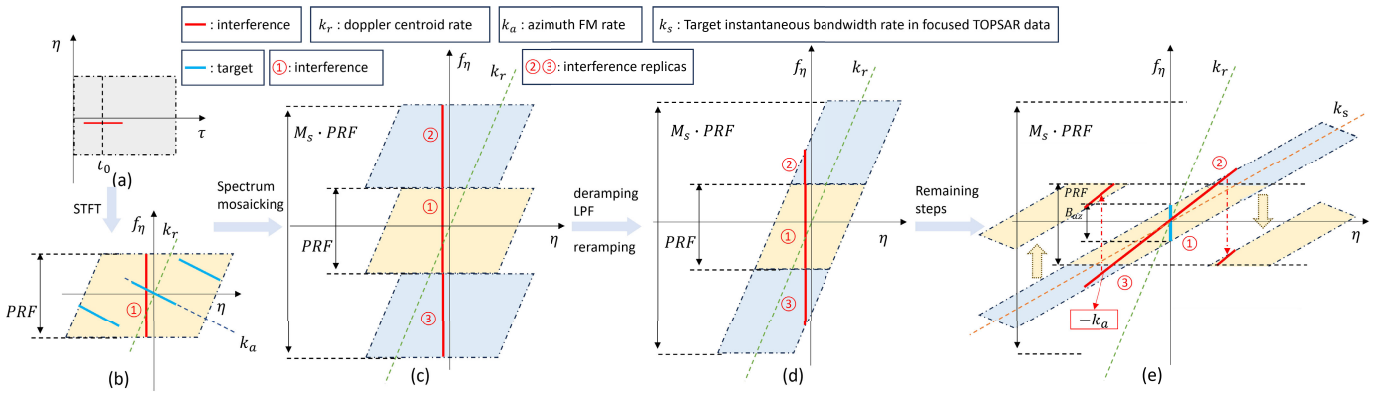


Fig. 3. Azimuth preprocessing (taking $M_s = 3$ as an example). (a) 2-D echo received by SAR. (b) Time–frequency spectrum of the azimuth signal at $t = t_0$. (c) After spectrum mosaicking. (d) After LPF. (e) After the remaining steps.

where f_a is the PRF, k_r is the Doppler centroid rate introduced by the scanning of the antenna, T_B is the burst duration, and $\lceil \cdot \rceil$ denotes the ceiling function.

Fig. 3 shows the effect of azimuth preprocessing on the signal model in Section II. Fig. 3(a) shows the 2-D echo matrix in the time domain. By fixing the time at $\tau = t_0$ and applying STFT to the corresponding azimuth signal, the result is shown in Fig. 3(b). Since the interference within this azimuth signal can be approximated as a δ function, its spectrum appears rectangular. Fig. 3(c) demonstrates the spectrum mosaicking process, where the generation of interference replicas can be observed. Due to the subsequent LPF, the bandwidth of each replica may be less than the PRF, as shown in Fig. 3(d).

Due to the limited support during azimuth focusing, time aliasing can also occur, so azimuth postprocessing is necessary [46]. After further processing, including azimuth focusing, RCMC, and azimuth postprocessing, the time–frequency distribution of the focused result is shown in Fig. 3(e). It can be observed that the original interference and its replicas suffer from frequency aliasing.

B. Impact of Azimuth Preprocessing on Interference Detection

After imaging, the target is focused into a sinc function. However, for targets at different azimuth times, the center of the Doppler bandwidth shifts with azimuth time at a rate k_s, which can be calculated as follows:

$$k_s = \frac{k_a \cdot k_r}{k_r + k_a}. \tag{20}$$

It should be noted that k_s is an equivalent mathematical parameter introduced to correct the time-domain aliasing. It represents the linear variation trend of the target's Doppler centroid in the focused data and does not correspond to any actual LFM signal component.

Based on the typical terrain observation by progressive scans SAR (TOPSAR) parameters provided in [46] and the phenomena observed in experiments, we have identified cases where there is little difference between $-k_a$ and k_s. Even though k_s does not represent an actual LFM component, we found that it still degrades detection performance, with a particularly sharp rise in the false alarm probability.

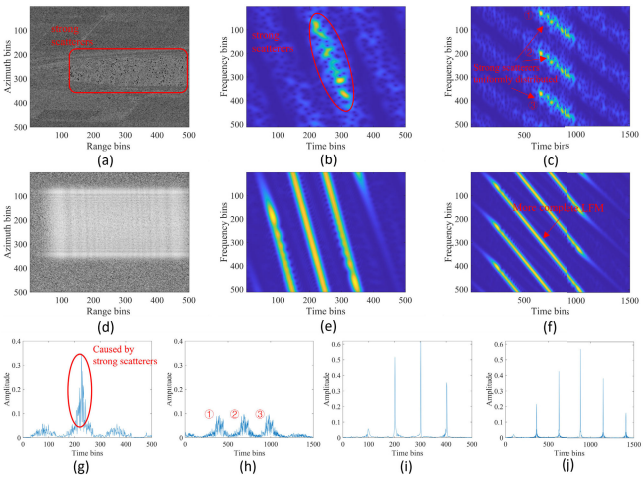


Fig. 4. Impact of azimuth preprocessing. (a) Clean image patch. (b) and (c) Time–frequency distributions of (a) without and with mosaicking, respectively. (d) Interfered image patch. (e) and (f) Time–frequency distributions of (d) without and with mosaicking, respectively. (g) and (h) FRFT results corresponding to (b) and (c). (i) and (j) FRFT results corresponding to (e) and (f).

Fig. 4(a) presents a 500×500 image patch extracted from the S-1 SLC data under TOPS mode. After performing SVD on it, Fig. 4(b) shows the time–frequency distribution of the first left singular vector. It can be observed that there are strong scatterers near the 300th azimuth bin, which is consistent with Fig. 4(a). Fig. 4(g) shows the FRFT result at the optimal angle, where periodic fluctuations of FRFT values can be observed. This characteristic is mainly attributed to the fact that the bandwidth center of the focused target retains the linear variation property of k_s, and periodic aliasing occurs when the target frequency exceeds the PRF. In addition, it can be observed that the presence of strong scatterers leads to a noticeable elevation of the FRFT values within the middle period, thereby increasing the false alarm probability and degrading detection performance.

Therefore, it is important to address the issue of FRFT peak elevation caused by strong scatterers. It is worth noting that this phenomenon is caused by isolated scatterers. If the scatterers completely cover the entire image patch, the energy

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within each period becomes uniformly distributed, and no significant peak will appear.

C. Interference Detection Preprocessing for TOPS Mode

According to Fig. 3, azimuth preprocessing introduces additional interference replicas. After imaging, these interference replicas also face periodic aliasing, yet they still exhibit phase continuity. However, since all targets are compressed into independent sinc functions, the targets within each period do not possess coherence. Based on this observation, we can perform mosaicking on the spectrum. First, perform a Fourier transform along each azimuth direction of patch $\mathbf{S}$ to obtain the range–Doppler domain signal $\mathbf{Y}(f)$

$$\mathbf{Y}(f) = \mathbf{F}_{\frac{\pi}{2}} \cdot \mathbf{S} \tag{21}$$

where $\mathbf{F}_{\pi/2}$ denotes the FRFT with an angle of $\pi/2$, i.e., the Fourier transform.

Then, perform spectrum mosaicking along the azimuth–Doppler axis

$$\hat{\mathbf{Y}}(f) = \mathbf{Y}(f - \beta f_a), \quad f \in \text{Band}_\beta \tag{22}$$

where $\beta = [-(M_s - 1)/2, \ldots, 0, \ldots, (M_s - 1)/2]$, $\text{Band}_\beta = \beta f_a + [-(f_a)/2, (f_a)/2]$, $\mathbf{Y}(f)$ is the original azimuth spectrum, and $\hat{\mathbf{Y}}(f)$ is the spliced spectrum.

Finally, perform the inverse Fourier transform along the azimuth–Doppler axis

$$\hat{\mathbf{S}} = \mathbf{F}_{\frac{\pi}{2}}^{-1} \cdot \hat{\mathbf{Y}}(f) \tag{23}$$

where $\mathbf{F}_{\pi/2}^{-1}$ denotes the inverse Fourier transform.

Fig. 4(c) shows the result of Fig. 4(b) processed with $M_s = 3$-fold mosaicking. It can be observed that mosaicking enables the strong scatterers to distribute periodically and uniformly. Since the energy of the singular vectors is fixed to one, the FRFT values induced by strong scatterers are equally distributed across all periods. As a result, the FRFT values shown in Fig. 4(h) still exhibit periodic fluctuations, but without localized extreme peaks. Compared to the prominent peaks observed in Fig. 4(g), the amplitude is reduced, thereby effectively reducing the false alarm probability.

With reference to Fig. 4, we analyze this from a mathematical perspective. Take an azimuth singular vector $u_1[n]$ of length N_a that contains isolated scatterers as an example; its energy is

$$E_1 = \sum_{n=0}^{N_a-1} |u_1[n]|^2 = 1. \tag{24}$$

Performing M times of spectrum mosaicking (in this example, $M = 3$) is mathematically equivalent to M-fold comb zero insertion in the time domain [47]. Therefore, the new sequence after mosaicking is expressed as

$$\hat{u}_1[m] = \begin{cases} u[m/M], & m = 0, M, 2M, \ldots, (N_a - 1)M \\ 0, & \text{else.} \end{cases} \tag{25}$$

Compared with $u_1[n]$, $\hat{u}_1[m]$ simply inserts periodic zero-valued points. Therefore, the total energy E_2 of $\hat{u}_1[m]$ remains unchanged, still equal to 1

$$E_2 = \sum_{m=0}^{MN_a-1} |\hat{u}_1[m]|^2 = \sum_{n=0}^{N_a-1} |\hat{u}_1[nM]|^2 = \sum_{n=0}^{N_a-1} |u_1[n]|^2 = 1. \tag{26}$$

To analyze the characteristics of $\hat{u}_1[m]$ in the fractional Fourier domain, we can express it as the product of $u_M[m]$ and $c[m]$, where $u_M[m]$ is the signal obtained by M-fold upsampling $u[n]$, its FRFT spectrum preserves the original energy focusing characteristic of $u_1[n]$ (i.e., a prominent isolated peak). $c[m]$ is a discrete impulse train with period M, defined as: it takes the value 1 every M points and 0 elsewhere. Using the discrete Fourier series, $c[m]$ can be expanded as [48]

$$c[m] = \frac{1}{M} \sum_{k=0}^{M-1} e^{j\frac{2\pi}{M}km}. \tag{27}$$

Therefore, $\hat{u}_1[m]$ can be expressed as

$$\hat{u}_1[m] = u_M[m] \cdot c[m] = \frac{1}{M} \sum_{k=0}^{M-1} u_M[m]\, e^{j\frac{2\pi}{M}km}. \tag{28}$$

Thus, $\hat{u}_1[m]$ can be seen as a superposition of M subterms. Taking the kth subterm as an example

$$\hat{u}_{1_k}[m] = \frac{1}{M} u_M[m] \cdot e^{j\frac{2\pi}{M}km}. \tag{29}$$

According to the modulation properties of the FRFT [49], the term $\exp(j\frac{2\pi}{M}km)$ induces a linear shift and phase modulation in the FRFT domain (along the u-axis). This means that the FRFT spectrum of each subterm is essentially a shifted version of the FRFT spectrum of $u_M[m]$. Furthermore, due to the linearity of the FRFT, applying the FRFT to the composite signal $\hat{u}_1[m]$ is equivalent to superimposing the FRFT spectra of all its constituent components, which essentially equates to shifting and superimposing the FRFT spectrum of $u_M[m]$. Since the total energy remains one, the originally highly concentrated energy at a single location is roughly uniformly distributed among the M shifted replica components, thereby significantly suppressing the extreme peak of the FRFT spectrum.

For interference, according to (22), spectrum mosaicking does not alter the original phase history of the signal. Therefore, the inherent quadratic phase structure and the chirp rate of the interference (i.e., the LFM signal) are strictly preserved. However, as mathematically analyzed above, this zero-insertion process also generates additional shifted replicas of the interference, which inherently tends to cause a reduction in the FRFT peak.

Nevertheless, as analyzed in Fig. 3, the original aliased replicas of the interference generated during imaging are strictly coherent with each other. Since the period of these replicas equals the PRF f_a, shifting and mosaicking the frequency-domain bands by integer multiples of f_a aligns these discontinuous aliased components. This reconstructs a continuous, wider-bandwidth LFM signal, thereby significantly alleviating the peak reduction or even enhancing the optimal FRFT peak value.

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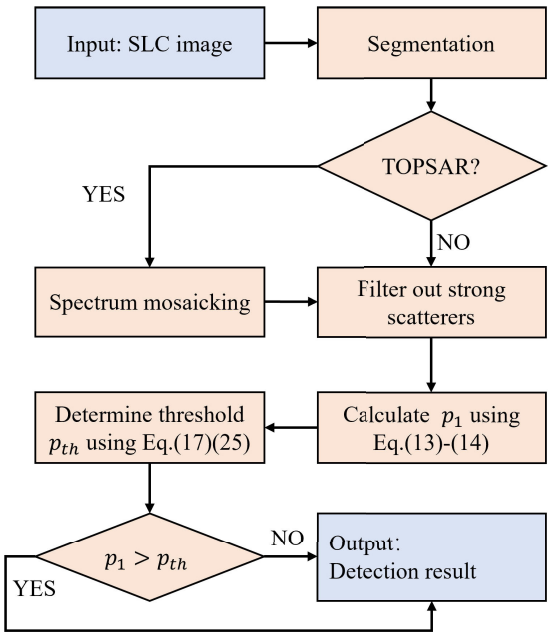


Fig. 5. Detection flow of SFID.

Fig. 4(e) and (f) present the time–frequency distributions without and with mosaicking, respectively. It can be seen that after mosaicking, more LFM replicas are generated, but a more complete LFM signal is formed. Fig. 4(i) and (j) correspond to the FRFT results at the optimal angle. It can be observed that the maximum peak does not decrease as severely as in Fig. 4(h), maintaining a relatively prominent peak. Through this processing, we can increase the difference in FRFT peak values between the H_0 and H_1 hypotheses, particularly by suppressing peaks caused by strong target clusters, thereby significantly reducing the false alarm probability.

D. Proof of Distribution Invariance

The key point that p_1 obeys (11) lies in the fact that SAR image pixels are i.i.d. and approximately follow a zero-mean complex Gaussian distribution; thus, their singular vector matrices are distributed according to the Haar measure on the unitary group. Since the Fourier transform and spectrum mosaicking do not alter the properties of the Gaussian distribution, $\hat{\mathbf{S}}$ still follows a zero-mean Gaussian distribution. Therefore, the derivation in Section III is still applicable to $\hat{\mathbf{S}}$, and its corresponding p_1 can be characterized by the following equation:

$$P\left(N_a' p_1 - 2\ln N_a' \le x\right) = \exp\left(-\exp\left(-x\right)\right) \tag{30}$$

where $N_a' = M_s \cdot N_a$.

V. Implementation Details

The complete SFID detection process is illustrated in Fig. 5, and the detection results can be directly fed into subsequent tasks such as interference suppression, interference parameter estimation, and interference localization. A detailed explanation of several essential steps is provided below.

A. Filter Out Strong Scatterers

SAR images may contain strong target scatterers such as vehicles [50], [51]. Since both the imaging operator and SVD are linear processes, the interference and strong target components are decoupled and do not interfere with one another.

Nevertheless, strong scatterers must be removed before detection for two reasons: first, for detection efficiency, this article only processes the first K singular vectors, but strong scatterers may dominate the leading singular vectors due to their large singular values, degrading detection performance; second, they cause the data distribution to deviate significantly from the Gaussian assumption, violating the theoretical prerequisites.

In summary, strong scatterers within an image patch should be filtered out in advance. In this article, simple processing is carried out using the following constant false alarm rate (CFAR) model:

$$\mathbf{D}\left[n,m\right] = \begin{cases} 1, & \mathbf{S}\left[n,m\right] < \text{mean}\left(|\mathbf{S}|\right) + \xi \cdot \text{std}\left(\mathbf{S}\right) \\ 0, & \mathbf{S}\left[n,m\right] \ge \text{mean}\left(|\mathbf{S}|\right) + \xi \cdot \text{std}\left(\mathbf{S}\right) \end{cases} \tag{31}$$

where mean(·) denotes the mean value, std(·) denotes the standard deviation, and ξ is the threshold coefficient of CFAR, which is set to five in this article. We then multiply the image patch $\mathbf{S}$ by $\mathbf{D}$ to filter out strong scatterers.

B. Azimuth-Overlapping Patch Processing

SFID relies on the abnormal peaks in the FRFT caused by the energy aggregation of LFM signals to detect interference. When the LFM signal is more complete in an image patch, its FRFT peak is theoretically higher, which is more conducive to interference detection. Therefore, when dividing the entire SAR image into patches, we introduce a certain degree of overlap between adjacent patches in the azimuth direction. This operation is also theoretically expected to enhance the detection performance of the Lambda-1 detector. To ensure the rigor of comparative experiments, we have also applied this overlapping operation to the Lambda-1 detector in the subsequent experimental section.

VI. Experimental Results

In this section, we conduct experiments on measured SLC images. Experiments 1 and 2 use clean images to verify false alarm performance, while also providing clean image patches for Experiment 6. Experiment 3 uses GF-3 images affected by dense interference. Experiments 4 and 5 use S-1 images with low ISR interference. In Experiment 6, we add interference with different ISR levels to clean image patches and plot the P_d and P_{fa} curves, thus validating the superiority of the proposed method over the traditional method.

A. Experiment 1: Gaofen-3 Clean Image

In the first experiment, a clean image from GF-3 is used. The image has dimensions of $6000 \times 20\,000$ pixels. It covers mostly sea areas, as well as land areas, including plains and urban areas. We set a window size of 500×500 pixels and

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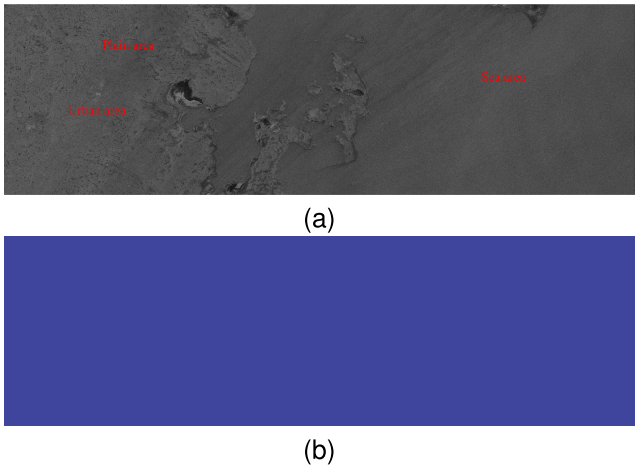


Fig. 6. Experimental image 1 and detection result. (a) Experimental image. (b) Detection result by SFID.

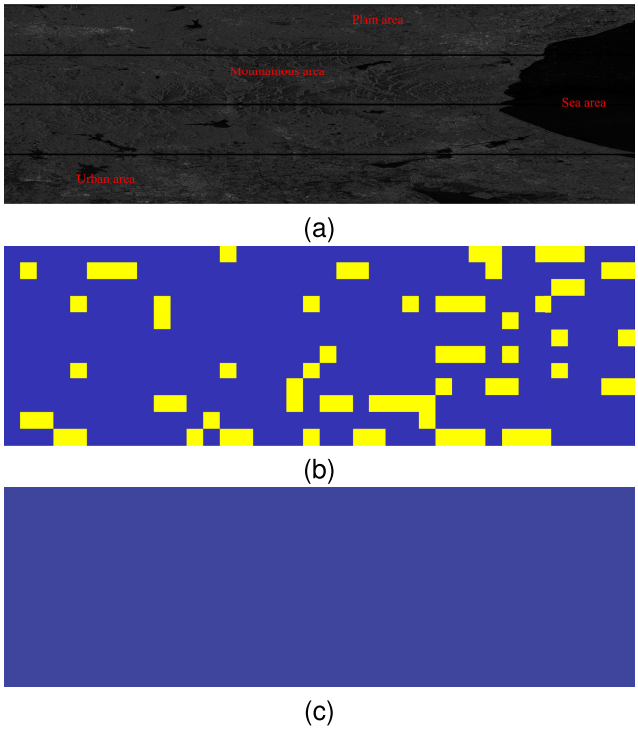


Fig. 7. Experimental image 2 and detection results. (a) Experimental image. (b) Detection results by SFID without preprocessing, and (c) by SFID with preprocessing.

divide the image into 12×40 patches. The detection result is shown in Fig. 6(b), where blue indicates that no interference was detected. In this experiment, the processing time is 15.6 s (using an i7-12700 CPU).

B. Experiment 2: Sentinel-1 Clean Image

In the second experiment, a clean image from S-1 is used. The image has dimensions of $6000 \times 19\,000$ pixels. This image covers most terrain types, such as sea areas, as well as land areas, including mountainous areas, plains, and urban areas, etc. It can provide abundant detection samples for Experiment 6, making the experimental results more convincing. We set a

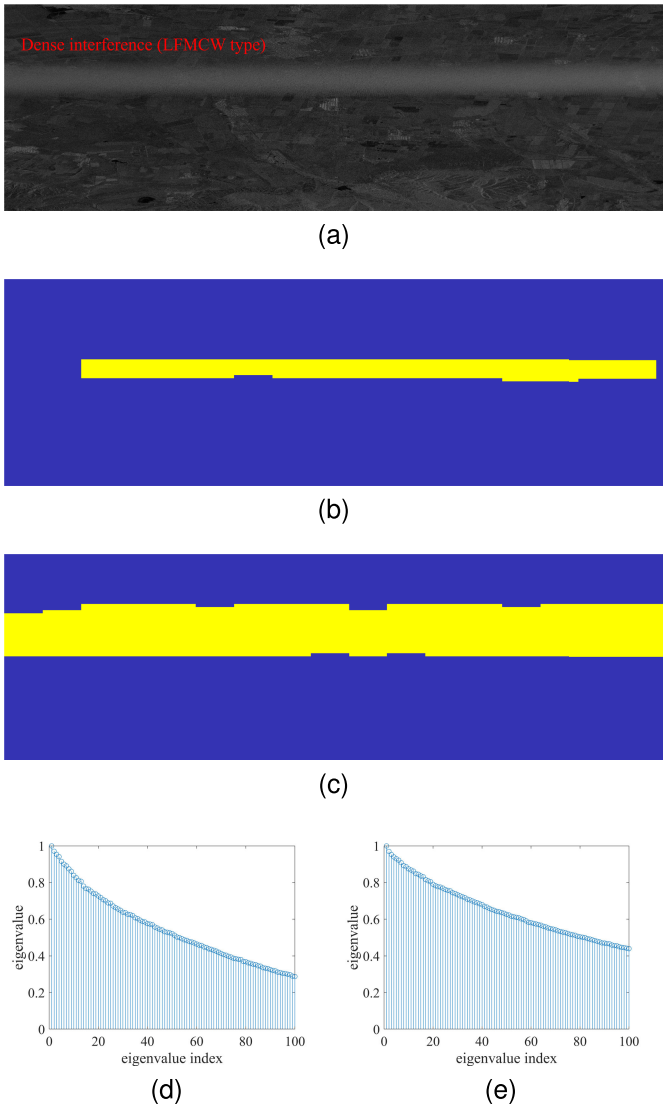


Fig. 8. Experimental image 3 and detection results. (a) Experimental image. (b) Detection results by the Lambda-1 detector, and (c) by SFID. (d) and (e) Eigenvalue distribution.

window size of 500×500 pixels and divide the image into 12×38 patches. First, without spectrum preprocessing, the patches are directly detected, and the result is shown in Fig. 7(b), where yellow indicates detected interference. It can be seen that some clean patches are detected as containing interference, leading to an increased false alarm probability. After applying preprocessing, the detection result is shown in Fig. 7(c). It is evident that the preprocessing effectively resolves the issue of increased false alarm probability caused by the TOPS mode. The processing time is 11.9 s.

C. Experiment 3: Gaofen-3 Image With Interference

In the third experiment, we used an interfered image from GF-3, as shown in Fig. 8(a). Analysis of its echo indicates that the interference is of the LFMCW type, with a high ISR and persistence across several PRI cycles, representing typical dense interference. The image dimensions are 6642×8595 pixels. We set a window size of 500×500

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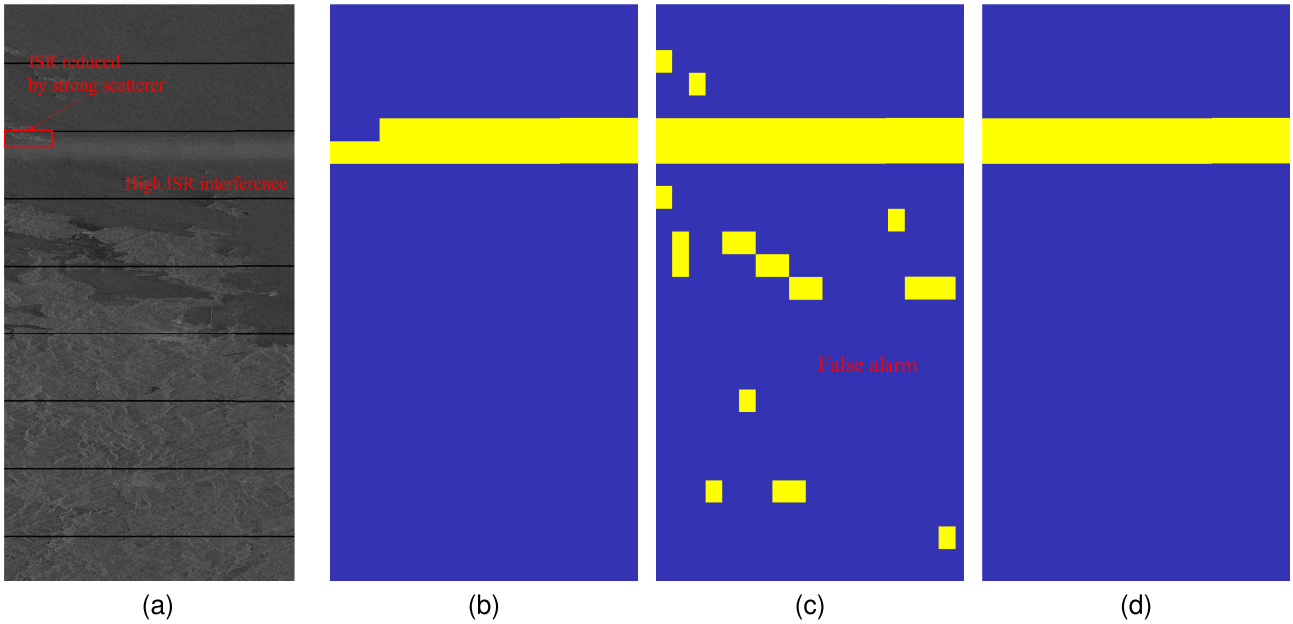


Fig. 9. Experimental image 4 and detection results. (a) Experimental image. (b) Detection results by Lambda-1 detector, (c) by SFID without preprocessing, and (d) by SFID with preprocessing.

pixels and an azimuth step of 100, dividing the image into 63 × 18 patches. The detection result from the Lambda-1 detector is shown in Fig. 8(b). It can be seen that this method can only detect the edge parts of the interference fringes, while most of the interfered areas are not detected. This is primarily because the undetected area is subject to dense interference, whereas the detected area has relatively sparse interference components, allowing its first eigenvalue to remain prominent. Fig. 8(d) shows the eigenvalue distribution of the detected area, and Fig. 8(e) shows the eigenvalue distribution of the undetected area that contains interference. It can be seen that, compared with Fig. 8(e), the eigenvalue decline trend in Fig. 8(d) is more pronounced, meaning that the first eigenvalue accounts for a higher proportion of energy relative to all eigenvalues. The detection result of the proposed SFID is shown in Fig. 8(c), which demonstrates that SFID handles dense interference more reliably compared with the traditional method. The processing time for this experiment is 56.7 s.

D. Experiment 4: Sentinel-1 Image With Interference

In the fourth experiment, we used an interfered image from S-1 (TOPS mode), which contains a high ISR interference, as shown in Fig. 9(a). The image dimensions are 12 720 × 9264 pixels. A window size of 500 × 500 pixels is applied, dividing the image into 25 × 18 patches. Fig. 9(b) presents the detection result of Lambda-1. Most interference over the sea surface is detected, though some remains undetected due to strong background regions in certain areas. Fig. 9(c) and (d), respectively, show the results with and without preprocessing. It can be observed that without detection preprocessing, SFID yields a large number of false alarm patches, which severely degrades the detection performance. In contrast, with hyperparameters γ kept unchanged, SFID successfully detects the interfered patches after preprocessing while maintaining a low false alarm probability. This demonstrates that the proposed preprocessing scheme for TOPS is feasible and effective. The processing time for this experiment is 17.3 s.

E. Experiment 5: Sentinel-1 Image With Interference

In the fifth experiment, an interfered image from S-1 is used (TOPS mode), as shown in Fig. 10(a). It can be observed that low ISR interference is present in the central region of the image. The image dimensions are 9922 × 8973 pixels. A window size of 500 × 500 pixels is set, dividing the image into 20 × 18 patches. The detection result of the Lambda-1 detector is shown in Fig. 10(b). The detection result of the proposed SFID is presented in Fig. 10(c) and (d). Without preprocessing, the false alarm probability is high. With hyperparameters γ kept unchanged, after preprocessing, the false alarm probability is significantly reduced, and SFID can effectively detect low ISR interference and dense interference. The processing time for this experiment is 10.84 s.

F. Analysis of P_{Fa} and P_d

The first five experiments validate the superiority of the proposed method from an image perspective. In this section, the false alarm probability P_{fa} and detection probability P_d are computed to quantitatively evaluate the method. Since the interference detection method in TOPS mode incorporates an additional preprocessing step compared to other imaging modes, two separate experiments, one for TOPS mode and one for StripMap mode, are conducted for rigorous comparison, though the experimental procedure remains the same. The experimental procedure is as follows: first, clean SAR images are divided to obtain N clean image patches. Then, by adjusting the hyperparameter γ, interference detection is performed to obtain P_{fa} under different γ values

$$P_{fa} = n_1/N \tag{32}$$

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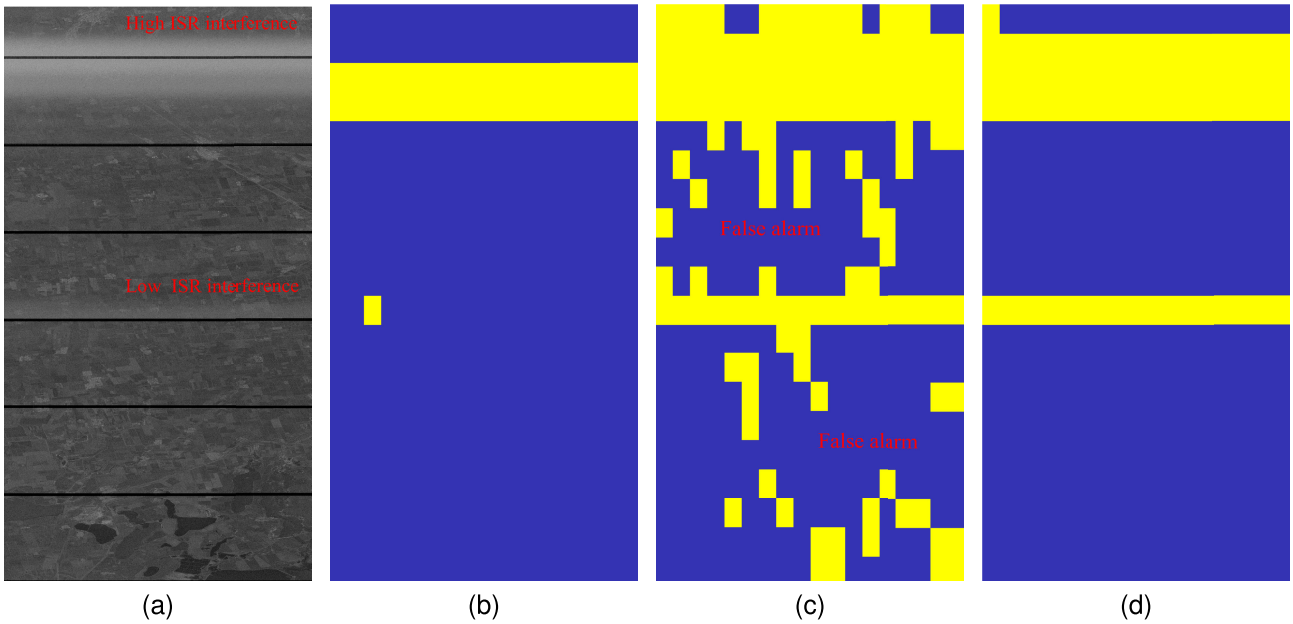


Fig. 10. Experimental image 5 and detection results. (a) Experimental image. (b) Detection results by Lambda-1 detector, (c) by SFID without preprocessing, and (d) by SFID with preprocessing.

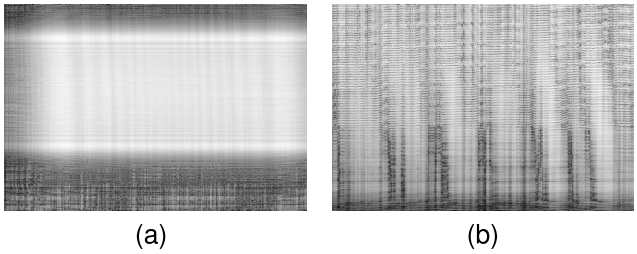


Fig. 11. (a) Extracted interference from S-1 image and (b) from GF-3 image.

where n_1 is the number of patches detected as containing interference. Subsequently, interference is added to the N clean patches, and both the ISR and γ are varied to obtain P_d under different parameter settings

$$P_d = n_2/N \tag{33}$$

where n_2 denotes the number of patches correctly detected as interfered.

The clean patches for TOPS mode and StripMap mode are directly taken from the images used in Experiments 1 and 2, with 480 and 456 patches, respectively. Interference signals extracted from other interfered images via the feature subspace separation method [5] are added to these clean patches. The extracted interference components are shown in Fig. 11.

Considering the differences between the interference to be added and the original clean images across experiments, the Lambda-1 detector is also tested under the same conditions. The results are shown in Figs. 12 and 13. In both TOPS and StripMap modes, when the ISR is not too low, the proposed method effectively detects interference with no false alarms occurring. However, when the ISR is low (e.g., −21 dB in the TOPS experiment and −18 dB in the StripMap experiment), the Lambda-1 detector fails to achieve reliable detection, whereas the proposed method still enables effective interference detection.

In summary, compared with the traditional SAR image detection method, the proposed SFID can effectively handle interference under low ISR conditions.

G. Discussion

1) Limitation Analysis: The proposed method exploits the phase information of the interference and is therefore applicable to SLC images; thus, it cannot detect interference in amplitude images. Furthermore, according to the results in Figs. 12 and 13. Compared with traditional methods, the proposed method can effectively detect weak interference. However, when the ISR decreases to a very low level (e.g., ISR < -21 dB in the S-1 experiment), the method may not perform well. Nevertheless, at such low ISR, the interference power is extremely low, its impact on the image is comparable to noise, and it does not seriously degrade image quality.

2) Advantages and Differences Compared With Other Methods: For traditional signal processing methods, including the Lambda-1 detector, and time-series-based methods, the proposed method deeply exploits the azimuthal structure information of interference in the image, thereby enabling better detection of low ISR interference. Moreover, it establishes a detection statistic model that obeys a fixed Gumbel distribution, allowing independent detection using only a single SLC image without requiring clean reference images or multipolarization channel images.

For deep learning-based methods, RFI typically manifests as bright stripes with distinct visual characteristics, making deep learning-based methods theoretically promising for detection. However, such methods generally require a large number of image samples covering various terrains and interference types

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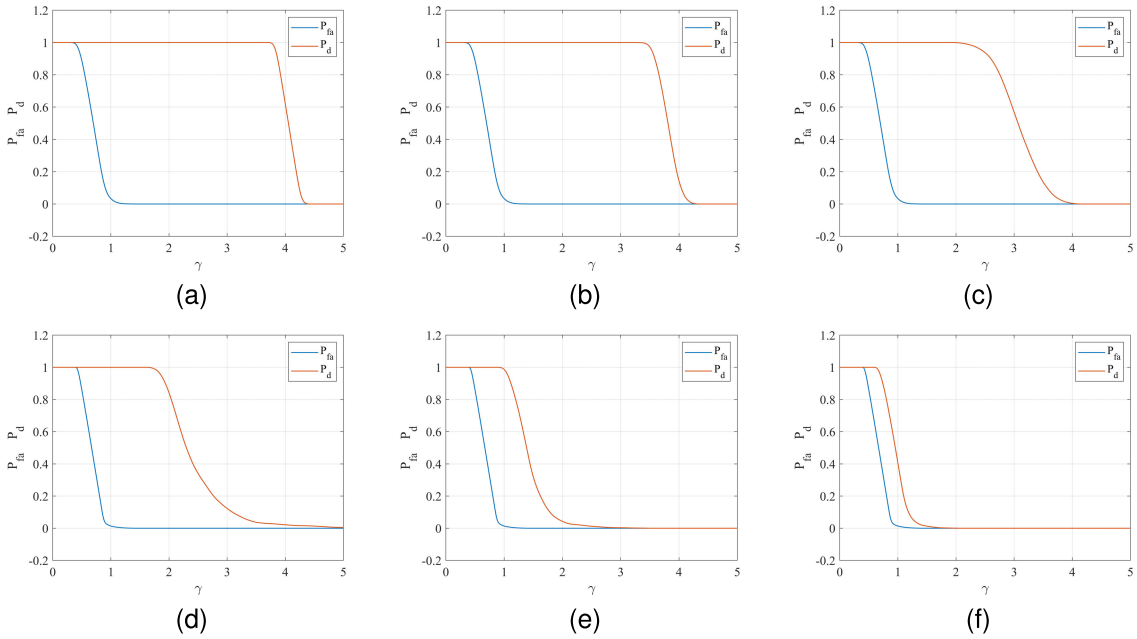


Fig. 12. S-1 experiment (TOPS): P_{fa} and P_d obtained under different threshold levels γp_{th} and ISR. (a)–(f) Results for SFID and Lambda-1 detector at −15,−18, and −21 dB, respectively.

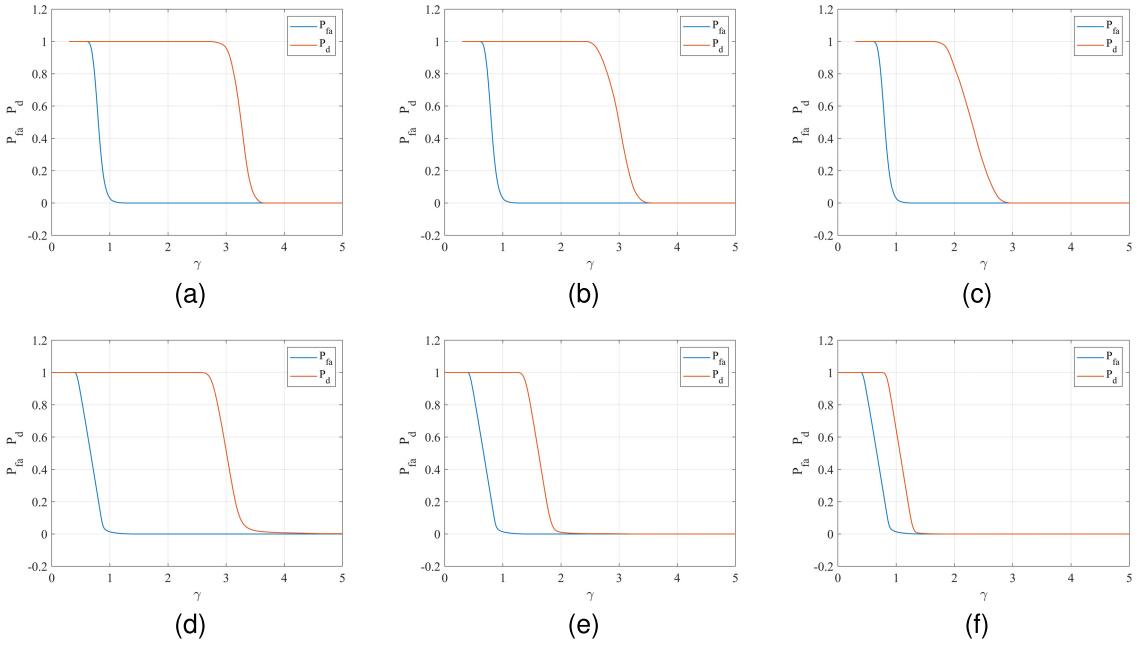


Fig. 13. GF-3 experiment (StripMap): P_{fa} and P_d obtained under different threshold levels γp_{th} and ISR. (a)–(f) Results for SFID and Lambda-1 detector at −10, −14, and −18 dB, respectively.

for training. Currently, most SAR RFI detection networks rely solely on data-driven approaches without exploiting the prior features of RFI, which may lead to misjudgements in scenarios not covered by the training set. In addition, existing methods typically take heavily compressed quick-look images as input [22], [25], aiming to exploit their small data volume to rapidly screen interfered images or locate interfered regions in quick-look images. If full-resolution uncompressed SLC images are used as input instead, significant computational overhead is often incurred. Meanwhile, directly mapping the detection results from the quick-look image to the actual full-resolution SLC image also introduces notable inaccuracies [52]. The reasons are twofold: first, the quick-look image is typically downsampled by a factor of several tens; second, taking the mainstream Sentinel-1 TOPS mode as an example, its SLC data consist of multiple "burst" subswaths, and there are invalid values between bursts, as shown in Fig. 10(a). In contrast, the quick-look image is usually a simplified display

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obtained by removing invalid values and stitching the bursts together. Both of these factors contribute to severe errors in direct mapping, rendering this strategy suitable only for coarse detection scenarios. In addition, due to differences in imaging parameters and preprocessing workflows among different SAR systems, the cross-system generalization capability of such methods also faces certain challenges [9].

However, as mentioned above, the proposed method does not apply to amplitude images, whereas deep learning-based methods can be well applied.

VII. Conclusion

This article proposes an adaptive interference detection method for SAR SLC images. The method mainly leverages the difference between the azimuth responses of targets and interference in focused images, where the interference exhibits an LFM characteristic. Specifically, this method performs SVD on each image patch and applies the FRFT at the optimal angle to the azimuth singular vector matrix. Interference detection is accomplished by comparing the squared global maximum value p_1 with a threshold. Theoretical derivation demonstrates that p_1 obeys the standard Gumbel distribution in clean images, thus enabling adaptive threshold determination based on this distribution. Due to the unique imaging mechanism of TOPSAR, this article analyzes and establishes an interference model for focused TOPSAR images and introduces a TOPSAR-specific detection preprocessing step. Experimental results show that, compared with the traditional SAR image interference detection method, the proposed method can effectively detect both low ISR interference and dense interference in various SAR images.

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References

[1] J. Su, M. Xi, Y. Gong, M. Tao, Y. Fan, and L. Wang, "Time-varying wideband interference mitigation for SAR via time-frequency-pulse joint decomposition algorithm," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5118816.

[2] Y. Mao et al., "A novel group-parametric model for RFI suppression on spaceborne SAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5201314.

[3] W. Fan, T. Tian, S. Wang, J. Du, X. Qi, and F. Zhou, "AMDIS-Net: A multiple attention mechanism based deep dense interference suppression network for Sentinel-1 SAR data," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 61, no. 6, pp. 1–19, Dec. 2025.

[4] Y. Huang, L. Zhang, J. Li, Z. Chen, and X. Yang, "Reweighted tensor factorization method for SAR narrowband and wideband interference mitigation using smoothing multiview tensor model," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3298–3313, May 2020.

[5] H. Yang, K. Li, J. Li, Y. Du, and J. Yang, "BSF: Block subspace filter for removing narrowband and wideband radio interference artifacts in single-look complex SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5211916.

[6] S. Wei, H. Zhang, X. Zeng, Z. Zhou, J. Shi, and X. Zhang, "CARNet: An effective method for SAR image interference suppression," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 114, Nov. 2022, Art. no. 103019.

[7] Y. Mao, Y. Huang, X. Yu, Y. Xin, Y. Wang, and W. Hong, "An radio frequency interference mitigation approach for spaceborne SAR system in low SINR condition," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5217414.

[8] S. Zhou et al., "Localization of ground-based periodic pulse interferers using time difference of arrival estimation in SAR satellite systems," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5204519.

[9] H. Yang, P. Lang, Y. He, X. Lu, Z. Liu, and J. Yang, "Lambda-1 detector: Adaptive interference detection in synthetic aperture radar images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5202613.

[10] X. Leng, K. Ji, and G. Kuang, "Radio frequency interference detection and localization in Sentinel-1 images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9270–9281, Nov. 2021.

[11] Z. Lv, H. Zhang, N. Li, and Z. Guo, "A two-step approach for pulse RFI detection in SAR data," in *Proc. IEEE Sensors*, Oct. 2021, pp. 1–4.

[12] N. Li, Z. Lv, and Z. Guo, "Observation and mitigation of mutual RFI between SAR satellites: A case study between Chinese GaoFen-3 and European Sentinel-1A," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5112819.

[13] T. Bollian, B. Osmanoglu, R. F. Rincon, S.-K. Lee, and T. E. Fatoyinbo, "Detection and geolocation of P-band radio frequency interference using EcoSAR," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 10, pp. 3608–3616, Oct. 2018.

[14] Z. Fu, T. Wei, N. Li, and N. Li, "Detection and localization of radio frequency interference via cross-domain multi-feature from SAR raw data," *J. Electron. Inf. Technol.*, vol. 47, no. 11, pp. 4351–4362, 2025.

[15] L. Yang, H. Zheng, J. Feng, N. Li, and J. Chen, "Detection and suppression of narrow band RFI for synthetic aperture radar imaging," *Chin. J. Aeronaut.*, vol. 28, no. 4, pp. 1189–1198, Aug. 2015. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1000936115001259

[16] B. Huang, H. Fattahi, H. Ghaemi, B. Hawkins, and G. Gunter, "Radio frequency interference detection and mitigation of NISAR data using slow time eigenvalue decomposition," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2023, pp. 5479–5482.

[17] R. Natsuaki, M. Jäger, and P. Prats-Iraola, "Similarity approach for radio frequency interference detection and correction in multi-receiver SAR," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Sep. 2020, pp. 3770–3773.

[18] A. Monti-Guarnieri, D. Giudici, and A. Recchia, "Identification of C-band radio frequency interferences from Sentinel-1 data," *Remote Sens.*, vol. 9, no. 11, p. 1183, Nov. 2017. [Online]. Available: https://www.mdpi.com/2072-4292/9/11/1183

[19] R. Zhang, L. Xu, Z. Yu, Y. Shi, C. Mu, and M. Xu, "Deep-IRTarget: An automatic target detector in infrared imagery using dual-domain feature extraction and allocation," *IEEE Trans. Multimedia*, vol. 24, pp. 1735–1749, 2022.

[20] X. Lu et al., "Automatic RFI identification for Sentinel-1 based on Siamese-type deep CNN using repeat-pass images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5231616.

[21] R. Zhang et al., "Differential feature awareness network within antagonistic learning for infrared-visible object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 8, pp. 6735–6748, Aug. 2024.

[22] J. Zhao et al., "Intelligent detection and segmentation of space-borne SAR radio frequency interference," *Remote Sens.*, vol. 15, no. 23, p. 5462, Nov. 2023.

[23] M. Tao et al., "Radio frequency interference signature detection in radar remote sensing image using semantic cognition enhancement network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5231714.

[24] X. Cen, Y. Li, Z. Han, T. Gu, P. Zhang, and T. Cai, "Self-supervised learning method for SAR multiinterference suppression," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5220017.

[25] A. Chojka, P. Artiemjew, and J. Rapiński, "RFI artefacts detection in Sentinel-1 level-1 SLC data based on image processing techniques," *Sensors*, vol. 20, no. 10, p. 2919, May 2020. [Online]. Available: https://www.mdpi.com/1424-8220/20/10/2919

[26] M. Tao, S. Tang, J. Li, X. Zhang, Y. Fan, and J. Su, "Radio frequency interference detection for SAR data using spectrogram-based semantic network," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, May 2021, pp. 1662–1665.

[27] J. Li, M. Tao, X. Zhang, J. Su, Y. Fan, and L. Wang, "A semantic cognition enhancment network for interference detection in Sentinel-1 SAR image," in *Proc. CIE Int. Conf. Radar (Radar)*, 2021, pp. 1923–1926.

[28] J. Yu, J. Li, B. Sun, J. Chen, and C. Li, "Multiclass radio frequency interference detection and suppression for SAR based on the single shot MultiBox detector," *Sensors*, vol. 18, no. 11, p. 4034, Nov. 2018.

[29] W. Fan et al., "Interference mitigation for synthetic aperture radar based on deep residual network," *Remote Sens.*, vol. 11, no. 14, p. 1654, Jul. 2019.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:03:22 UTC from IEEE Xplore. Restrictions apply.

[30] M. Tao, S. Lai, J. Li, J. Su, Y. Fan, and L. Wang, "Extraction and mitigation of radio frequency interference artifacts based on time-series Sentinel-1 SAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5217211.

[31] N. Li, H. Zhang, Z. Lv, L. Min, and Z. Guo, "Simultaneous screening and detection of RFI from massive SAR images: A case study on European Sentinel-1," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5231917.

[32] R. Zhang et al., "A benchmark and frequency compression method for infrared few-shot object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5001711.

[33] J. Zhang, R. Zhang, Z. Cao, L. Xu, X. Chen, and M. Xu, "It takes two: Multi-frequency perception with complementary fusion network for complex scene segmentation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 36, no. 4, pp. 5288–5300, Apr. 2026.

[34] I. G. Cumming and F. H. Wong, *Digital Processing of Synthetic Aperture Radar Data: Algorithms and Implementation*. Norwood, MA, USA: Artech House, 2005.

[35] G. H. Golub and C. F. Van Loan, *Matrix Computations*, 4th ed., Philadelphia, PA, USA: Johns Hopkins University Press, 2013. [Online]. Available: https://epubs.siam.org/doi/abs/10.1137/1.9781421407944

[36] A. Edelman and N. R. Rao, "Random matrix theory," *Acta Numerica*, vol. 14, pp. 233–297, Jun. 2005.

[37] T. Jiang, "Maxima of entries of Haar distributed matrices," *Probab. Theory Rel. Fields*, vol. 131, no. 1, pp. 121–144, Jan. 2005.

[38] A. Haar, "Der massbegriff in der theorie der kontinuierlichen gruppen," *Ann. Math.*, vol. 34, no. 1, pp. 147–169, Jan. 1933. [Online]. Available: http://www.jstor.org/stable/1968346

[39] J. W. Silverstein, "Describing the behavior of eigenvectors of random matrices using sequences of measures on orthogonal groups," *SIAM J. Math. Anal.*, vol. 12, no. 2, pp. 274–281, Mar. 1981, doi: 10.1137/0512025.

[40] Z. D. Bai, B. Q. Miao, and G. M. Pan, "On asymptotics of eigenvectors of large sample covariance matrix," *Ann. Probab.*, vol. 35, no. 4, pp. 1532–1572, Jul. 2007, doi: 10.1214/009117906000001079.

[41] Z. D. Bai, H. X. Liu, and W. K. Wong, "Asymptotic properties of eigenmatrices of a large sample covariance matrix," *Ann. Appl. Probab.*, vol. 21, no. 5, pp. 1994–2015, Oct. 2011, doi: 10.1214/10-aap748.

[42] Y. Yin and W. Zhou, "Limiting behavior of bilinear forms for the resolvent of sample covariance matrices under elliptical distribution with applications," 2023, *arXiv:2312.16373*.

[43] A. Knowles and J. Yin, "Anisotropic local laws for random matrices," *Probab. Theory Rel. Fields*, vol. 169, nos. 1–2, pp. 257–352, Oct. 2017.

[44] N. S. Pillai and J. Yin, "Universality of covariance matrices," *Ann. Appl. Probab.*, vol. 24, no. 3, pp. 935–1001, Jun. 2014.

[45] O. Aldimashki and A. Serbes, "LFM signal parameter estimation in the fractional Fourier domains: Analytical models and a high-performance algorithm," *Signal Process.*, vol. 214, Jan. 2024, Art. no. 109224. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0165168423002980

[46] F. De Zan and A. M. Guarnieri, "TOPSAR: Terrain observation by progressive scans," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 9, pp. 2352–2360, Sep. 2006.

[47] A. Oppenheim and R. Schafer, *Discrete-time Signal Processing*. London, U.K.: Pearson, 2010. [Online]. Available: https://books.google.com.sg/books?id=mYsoAQAAMAAJ

[48] A. Oppenheim, A. Willsky, and S. Nawab, *Signals & Systems*. Upper Saddle River, NJ, USA: Prentice-Hall, 1997.

[49] H. M. Ozaktas, M. A. Kutay, and D. Mendlovic, "Introduction to the fractional Fourier transform and its applications," in *Advances in Imaging and Electron Physics*. Amsterdam, The Netherlands: Elsevier, 1999, pp. 239–291. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1076567008702726

[50] D. Guan, R. Feng, Y. Xie, H. Ding, Y. Cui, and D. Xiang, "SAR vehicle data generation with scattering features for target recognition," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 5520–5538, 2026.

[51] Y. Xie, H. Ding, Y. Liu, N. Li, D. Xiang, and Y. Su, "AZI-VEC: High-resolution azimuth-sensitive PolSAR image generation for vehicle automatic target recognition," *ISPRS J. Photogramm. Remote Sens.*, vol. 236, pp. 65–80, Jun. 2026. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0924271626001528

[52] J. Shen et al., "A radio frequency interference screening framework—From quick-look detection using statistics-assisted network to raw echo tracing," *Remote Sens.*, vol. 16, no. 22, p. 4195, Nov. 2024. [Online]. Available: https://www.mdpi.com/2072-4292/16/22/4195

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